

Integration — Lesson 4 Test: The Antiderivative — Undoing the Derivative

10 questions · show your work in the box · check every antiderivative by differentiating it · don't forget + C on family answers

Name: _____ Date: _____ Score: _____ / 10

1. Find the antiderivative of $8x$. (Give the whole family of answers.)

1 pt

2. Find the antiderivative of 6 (just the constant 6).

1 pt

3. Find the antiderivative of $5x^4$, then check your answer by differentiating it. Show the check.

1 pt

4. Find the antiderivative of $4x + 6$.

1 pt

5. Check whether $x^3 + 2x$ is an antiderivative of $3x^2 + 2$. Show the differentiation that proves your answer. ^{1 pt}

6. Check whether $x^2 + x$ is an antiderivative of $2x$. If it is not, say exactly what goes wrong. ^{1 pt}

7. A student writes: "The antiderivative of $6x$ is $3x^2 + 100$." Their classmate says that must be wrong because the book's answer is $3x^2 + C$. Who is right? Explain using a derivative check. ^{1 pt}

- 8.** Explain in one or two sentences why every antiderivative answer ends in "+ C". (What happens to constants when you differentiate?) 1 pt

- 9.** A function f has derivative $f'(x) = 2x$, and $f(2) = 7$. Find the exact function $f(x)$ — no C should remain in your final answer. 1 pt

- 10.** Find $f(x)$ given $f'(x) = 6x + 1$ and $f(1) = 9$. Then verify both given facts about your answer. 1 pt
